A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 1: Information Representation

Subtopic 1.1: Data Representation

Concept 1.1.2: Number Systems and Conversions (Binary, Denary, Hexadecimal)

Total Questions: 25

Mark Schemes begin on Page 11

Question 3 | Source: 9618_w25_qp_11 (Q1.a.i)

1 **(a)** **(i)** Convert the binary number into hexadecimal.

101100111010

.... [1]

Question 4 | Source: 9618_w25_qp_12 (Q6.b)

(b) The following table shows some words and corresponding denary values.

Word	Denary value
Computing	55
Science	56
Computers	57
are	58
Brilliant!	59
is	60
Fun!	61
Amazing!	62

The following table shows three bytes of data that have been received.

Use the table to find the corresponding words from the binary values received.

Binary value	00111000	00111100	00111110
Word			

Working ....

....

....

....

[1]

Question 5 | Source: 9618_w25_qp_12 (Q7.c.ii)

(c) ASCII is another character set. The ASCII value for the character ‘h’ has the denary value 104.

(ii) Write the hexadecimal value for the ASCII character ‘h’.

.... [1]

Question 6 | Source: 9618_s25_qp_11 (Q3.b.ii)

3 A computer stores images and text files.

(b) The text in the files is stored using the Unicode character set.

(ii) The Unicode character '**<**' has the binary value: 0010 0111 0110 1110

Convert the binary value for the character '**<**' into denary.

.... [1]

Question 7 | Source: 9618_s25_qp_11 (Q3.b.iii)

3 A computer stores images and text files.

(b) The text in the files is stored using the Unicode character set.

(iii) The Unicode character '∑' has the hexadecimal value 2140

Convert the hexadecimal code for the character '∑' into denary.

.... [1]

Question 8 | Source: 9618_s25_qp_12 (Q2.a)

2 **(a)** Convert the denary integer 558 into 12-bit binary and hexadecimal.

Binary ....

Hexadecimal ....

[2]

Question 9 | Source: 9618_w24_qp_11 (Q1.b.i)

(b) **(i)** Convert the unsigned binary integer into hexadecimal.

110001100111

Answer .... [1]

Question 10 | Source: 9618_w24_qp_13 (Q8.a)

8 **(a)** Convert the hexadecimal number 1FAB into denary.

Working ....

....

....

Denary value .... [1]

Question 11 | Source: 9618_s24_qp_12 (Q7.c)

7 A computer stores binary data.

(c) Convert the hexadecimal number C0F into denary.

Show your working.

Working ....

....

....

Answer ....

[2]

Question 12 | Source: 9618_s24_qp_13 (Q1.b)

(b) Convert the denary number 241 to hexadecimal.

....

.... [1]

Question 13 | Source: 9618_w23_qp_12 (Q3.b.iii)

(iii) Convert the unsigned binary integer into denary.

000111010110

Answer .... [1]

Question 14 | Source: 9618_s23_qp_11 (Q3.d.iii)

3 A computer has an Operating System (OS).

(d) The computer stores data in binary form.

(iii) Convert the positive binary integer 11110010 into hexadecimal.

....

.... [1]

Question 15 | Source: 9618_s23_qp_12 (Q4.a)

4 Data in a computer is stored in binary form.

(a) State the number of unique binary values that can be represented in 16 bits.

.... [1]

Question 16 | Source: 9618_s23_qp_12 (Q4.c)

4 Data in a computer is stored in binary form.

(c) Convert the hexadecimal number A04 into denary.

Show your working.

Working ....

....

....

....

Answer ....
[2]

Question 17 | Source: 9618_w22_qp_11 (Q1.a.i)

1 **(a)** **(i)** Convert the unsigned binary integer into denary.

00100111

Answer .... [1]

Question 18 | Source: 9618_w22_qp_12 (Q2.a.ii)

(ii) Convert the unsigned binary integer into hexadecimal.

10010110

Answer .... [1]

Question 19 | Source: 9618_w22_qp_13 (Q9.a.i)

9 **(a)** **(i)** Convert the unsigned binary value into hexadecimal.

10010011

Answer .... [1]

Question 20 | Source: 9618_w22_qp_13 (Q9.a.ii)

(ii) Convert the unsigned binary value into denary.

10010011

Answer .... [1]

Question 21 | Source: 9618_w21_qp_11 (Q1.c)

(c) Convert the hexadecimal value F0 into denary.

....

.... [1]

Question 22 | Source: 9618_w21_qp_12 (Q4.a)

4 A register stores the following binary number:

1	1	0	0	1	1	0	1

(a) The binary value in the register represents an unsigned binary integer.

Convert the unsigned binary integer into denary.

.... [1]

Question 23 | Source: 9618_w21_qp_12 (Q4.c)

4 A register stores the following binary number:

1	1	0	0	1	1	0	1

(c) The binary value in the register represents a hexadecimal number.

Convert the binary number into hexadecimal.

.... [1]

Question 24 | Source: 9618_s21_qp_11 (Q1.c.i)

1 Anya scans an image into her computer for a school project.

(c) One of the colours used in the image has the hexadecimal colour code:

`#FC238A`

`FC` is the amount of red, `23` is the amount of green and `8A` is the amount of blue in the colour.

(i) Convert the hexadecimal code `FC` into denary.

.... [1]

Question 25 | Source: 9618_s21_qp_12 (Q4.b.i)

4 The table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		**Explanation**
Opcode	**Operand**	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
STO	<address>	Store contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX)
CMP	<address>	Compare the contents of ACC with the contents of <address>
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
JMP	<address>	Jump to the given address
IN		Key in a character and store its ASCII value in ACC
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system
# denotes a denary number, e.g. #123		

The current contents of the main memory and selected values from the ASCII character set are:

Address	**Instruction**
70	IN
71	CMP 100
72	JPE 80
73	CMP 101
74	JPE 76
75	JMP 80
76	LDD 102
77	INC ACC
78	STO 102
79	JMP 70
80	LDD 102
81	DEC ACC
82	STO 102
83	JMP 70
…	
100	68
101	65
102	100

ASCII code table (selected codes only)

ASCII code	**Character**
65	A
66	B
67	C
68	D

(b) Some bit manipulation instructions are shown in the table:

Instruction		Explanation
Opcode	**Operand**	
AND	#n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
<address> can be an absolute address or a symbolic address # denotes a denary number, e.g. #123		

The contents of the memory address 300 are shown:

Bit Number	7	6	5	4	3	2	1	0
300	0	1	1	0	0	1	1	0

(i) The contents of memory address 300 represent an unsigned binary integer.

Write the denary value of the unsigned binary integer in memory address 300.

.... [1]

Question 26 | Source: 9618_s21_qp_12 (Q6.c.i)

6 A computer uses the ASCII character set.

(c) Unicode is a different character set.

The Unicode value for the character ‘1’ is denary value 49.

(i) Write the hexadecimal value for the Unicode character ‘1’.

.... [1]

Question 27 | Source: 9618_s21_qp_12 (Q6.c.ii)

6 A computer uses the ASCII character set.

(c) Unicode is a different character set.

The Unicode value for the character ‘1’ is denary value 49.

(ii) Write the denary value for the Unicode character ‘5’.

.... [1]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_13 (Q2.a)

2(a)

3 marks for 6 correct ticks
2 mark for 4 or 5 correct ticks
1 mark for 2 or 3 correct ticks

Example of data	Number of bits						
	4	8	16	24	32	64	128
the hexadecimal value F139			✓				
16 000 000 unique amplitude values				✓			
an IPv4 address					✓		
256 unique colours		✓					
an IPv6 address							✓
the denary value 65 000			✓				

Marks: 3

Mark Scheme for Question 2 | Source: 9618_w25_ms_13 (Q2.d)

2(d)	**1 mark** E33B	1

Mark Scheme for Question 3 | Source: 9618_w25_ms_11 (Q1.a.i)

1(a)(i)	B3A	1

Mark Scheme for Question 4 | Source: 9618_w25_ms_12 (Q6.b)

6(b)

1 mark for correct words in the correct order

Binary Value	0011 1000	0011 1100	0011 1110
Word	Science	is	Amazing!

Marks: 1

Mark Scheme for Question 5 | Source: 9618_w25_ms_12 (Q7.c.ii)

7(c)(ii)	**1 mark** for the correct answer 68	**1**

Mark Scheme for Question 6 | Source: 9618_s25_ms_11 (Q3.b.ii)

3(b)(ii)	**1 mark** for: 10,094	**1**

Mark Scheme for Question 7 | Source: 9618_s25_ms_11 (Q3.b.iii)

3(b)(iii)	**1 mark** for: 8,512	**1**

Mark Scheme for Question 8 | Source: 9618_s25_ms_12 (Q2.a)

2(a)	**1 mark** for: 0010 0010 1110 **1 mark** for: 22E	**2**

Mark Scheme for Question 9 | Source: 9618_w24_ms_11 (Q1.b.i)

1(b)(i)	**1 mark** for: C67	**1**

Mark Scheme for Question 10 | Source: 9618_w24_ms_13 (Q8.a)

8(a)	**1 mark** for: Denary value: 8107	**1**

Mark Scheme for Question 11 | Source: 9618_s24_ms_12 (Q7.c)

7(c)	**1 mark** for working: 1100 0000 1111 // 2048 + 1024 + 8 + 4 + 2 + 1 // (12 * 16^2) + 15 // (12 * 16 * 16) + 15 // 3072 + 15 **1 mark** for correct answer: 3087	**2**

Mark Scheme for Question 12 | Source: 9618_s24_ms_13 (Q1.b)

1(b)	**1 mark** for correct answer: **F1**	**1**

Mark Scheme for Question 13 | Source: 9618_w23_ms_12 (Q3.b.iii)

3(b)(iii)	470	**1**

Mark Scheme for Question 14 | Source: 9618_s23_ms_11 (Q3.d.iii)

3(d)(iii)	F2	**1**

Mark Scheme for Question 15 | Source: 9618_s23_ms_12 (Q4.a)

4(a)	2^{16} // 65536	**1**

Mark Scheme for Question 16 | Source: 9618_s23_ms_12 (Q4.c)

4(c)	**1 mark** for working; **1 mark** for answer
	
	• Working: A04 = (10 * 16^2) + 4
	// A04 = (10 * 256) + 4
	// A04 = 1010 0000 0100
	
	• Answer: 2564
		2

Mark Scheme for Question 17 | Source: 9618_w22_ms_11 (Q1.a.i)

1(a)(i)	39	**1**

Mark Scheme for Question 18 | Source: 9618_w22_ms_12 (Q2.a.ii)

2(a)(ii)	96	**1**

Mark Scheme for Question 19 | Source: 9618_w22_ms_13 (Q9.a.i)

9(a)(i)	93	**1**

Mark Scheme for Question 20 | Source: 9618_w22_ms_13 (Q9.a.ii)

9(a)(ii)	147	**1**

Mark Scheme for Question 21 | Source: 9618_w21_ms_11 (Q1.c)

1(c)	240	**1**

Mark Scheme for Question 22 | Source: 9618_w21_ms_12 (Q4.a)

4(a)	205	**1**

Mark Scheme for Question 23 | Source: 9618_w21_ms_12 (Q4.c)

4(c)	CD	**1**

Mark Scheme for Question 24 | Source: 9618_s21_ms_11 (Q1.c.i)

1(c)(i)	252	**1**

Mark Scheme for Question 25 | Source: 9618_s21_ms_12 (Q4.b.i)

4(b)(i)	102	**1**

Mark Scheme for Question 26 | Source: 9618_s21_ms_12 (Q6.c.i)

6(c)(i)	31	**1**

Mark Scheme for Question 27 | Source: 9618_s21_ms_12 (Q6.c.ii)

6(c)(ii)	53	**1**